

The 3×3 Magic Square of Squares: Structure, Conditions, Eigenmatrix, and Value Limits

Shaimaa Soltan

August 2026

Abstract

A summary of the algebraic structure of the 3×3 magic square, reduced to its two shape parameters, together with the conditions for nine distinct cells, the modular sieves on the offsets, the eigen-structure of the offset matrix $\mathbf{N}$, and the numerical limits on all the parameters. The governing object is a single shape ratio $r = C/A$ and an offset matrix $\mathbf{N} = a\mathbf{P} + c\mathbf{Q}$.

1 Setup and parametrization

A 3×3 magic square is written row-major as

$$\begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}, \quad \text{all three rows, three columns and both diagonals equal the sum } S.$$

Center law. Every line through the center forces

$$\boxed{E = \frac{S}{3}} \quad \Longleftrightarrow \quad S = 3E,$$

because the two cells flanking the center are $E \mp (A - C)$, whose offsets cancel. Equivalently, each cell plus its point-reflection through the center satisfies

$$A + I = B + H = C + G = D + F = 2E \quad (\text{opposite cells sum to } 2E).$$

Generator basis. Subtracting the center and factoring gives the *offset matrix*

$$\mathbf{N} = a\mathbf{P} + c\mathbf{Q}, \quad \mathbf{P} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix},$$

$$\mathbf{N} = \begin{pmatrix} a & -a - c & c \\ -a + c & 0 & a - c \\ -c & a + c & -a \end{pmatrix},$$

where $\mathbf{P} = \mathbf{P}^\top$ is symmetric (the “ a ” generator) and $\mathbf{Q} = -\mathbf{Q}^\top$ is skew-symmetric (the “ c ” generator). Each of $\mathbf{P}, \mathbf{Q}$ has every line summing to 0.

Full square. With $\mathbf{J}$ the all-ones matrix and center scale E ,

$$\boxed{\text{Square} = E(\mathbf{J} + \mathbf{N})}, \quad \text{cell} = E(1 + n), \quad n \in \{0, \pm a, \pm c, \pm(a + c), \pm(a - c)\}.$$

2 Magic conditions and degrees of freedom

Object	Independent line-equations	Degrees of freedom
Full magic square	7 (of 8; one is dependent)	$9 - 7 = 2$ (plus scale E)
Semi-magic (rows+cols)	5	$9 - 5 = 4$

The single dependency among the eight lines is the grand total: $R_1 + R_2 + R_3 = (\text{all cells}) = C_1 + C_2 + C_3$. In the normalized variables the two non-trivial magic conditions are

$$a + b + c = 0, \quad a + d - c = 0 \quad \implies \quad b = -a - c, \quad d = c - a,$$

leaving exactly the two free parameters a and c .

3 Normalized forms

$$\underbrace{\text{cells}}_{\text{sum } 3E} \xrightarrow{\div E} \mathbf{M} = \mathbf{J} + \mathbf{N} \quad (\text{center } 1, \text{ sum } 3) \xrightarrow{\div 3} \frac{1}{3}\mathbf{M} \quad (\text{center } \frac{1}{3}, \text{ sum } 1).$$

$$\mathbf{M} = \begin{pmatrix} 1 + a & 1 - a - c & 1 + c \\ 1 - a + c & 1 & 1 + a - c \\ 1 - c & 1 + a + c & 1 - a \end{pmatrix}.$$

- **Center-1 form** ($\div E$): *square-friendly*. Since $E = \rho^2$ is a perfect square, an integer-square cell $\Leftrightarrow$ a rational-square entry of $\mathbf{M}$. Use this frame for the square hunt.
- **Sum-1 form** ($\div 3E$): *barycentric*. Center = $\frac{1}{3}$, each flanking pair = $\frac{2}{3}$. Do *not* use for squares: dividing by 3 (not a square) breaks the integer $\leftrightarrow$ rational-square correspondence.

4 Reduced problem: shape ratio and forbidden values

Factor a from $\mathbf{N}$ and set the shape ratio

$$\boxed{r = \frac{c}{a} = \frac{C}{A}}, \quad \mathbf{N} = a(\mathbf{P} + r\mathbf{Q}).$$

The four offset magnitudes are proportional to 1, r , $1 + r$, $1 - r$ (directions $0^\circ, 90^\circ, 45^\circ, 135^\circ$ in the (a, c) plane).

Symmetry. The 8 magic-square symmetries act on the ratio as $r \mapsto \{-r, 1/r, -1/r\}$, so every distinct shape has a representative in

$$\boxed{0 < r < 1} \quad (\text{fundamental domain}).$$

Forbidden ratios (no 9 distinct cells). A cell duplicates or hits the center exactly when

$$\boxed{r \in \{0, \pm \frac{1}{2}, \pm 1, \pm 2, \infty\}}$$

i.e. the axes $(0, \infty)$, the diagonals (± 1) , and their 45° reflection partners $(\pm \frac{1}{2}, \pm 2)$. In the fundamental range the only forbidden value is $r = \frac{1}{2}$ (endpoints 0, 1 excluded).

Positivity (all nine cells > 0).

$$|a| < \frac{1}{\max(1, |r|, |1+r|, |1-r|)} = \frac{1}{1+r} \quad (0 < r < 1).$$

This bounds the offset scale a ; it does *not* bound r .

Two kinds of boundary (radius vs. direction). Read (r, a) as polar coordinates: r is the *direction* and a is the *radius* along it. The two limits act on different coordinates:

- The **forbidden ratios** delete whole *directions* (rays through the origin): $r \in \{0, \pm\frac{1}{2}, \pm 1, \pm 2, \infty\}$. This is a limit *on* r —it removes values of the shape.
- **Positivity** caps the *radius*: $|a| < 1/(1+r)$. This is a limit *on* a . Its numerical value is a *function* of r , but it never deletes a direction: every admissible $r \in (0, 1)$ keeps a nonempty window $|a| < 1/(1+r)$. A direction-dependent cap on the radius is not a restriction on the direction.

Thus “bounds a , not r ” means exactly: positivity excludes no value of r ; it only fixes how large $|a|$ may be once the direction r is chosen. The tightest cell is $1 - a(1+r)$ (largest magnitude $1+r$ on $0 < r < 1$), which is why the cap is $1/(1+r)$.

Geometry in the (a, c) -plane. Positivity is the open hexagon about the origin

$$\max(|a|, |c|, |a+c|, |a-c|) < 1,$$

and the forbidden ratios are the finite set of *rays* $c = ra$, $r \in \{0, \pm\frac{1}{2}, \pm 1, \pm 2, \infty\}$, deleted from it. The hexagon caps the radius (the a -bound, tighter in some directions); the deleted rays cut out directions (the r -limit)—two different boundaries on the same 2-dimensional shape region.

5 Square conditions and the elliptic curve

Nine cells are perfect squares iff, in the center-1 frame,

$$1 \pm a, \quad 1 \pm c, \quad 1 \pm (a+c), \quad 1 \pm (a-c)$$

are all rational squares (the center 1 is free). Each pair “ $1 \pm x$ both squares” is a rational point on the conic $p^2 + q^2 = 2$ (genus 0, $\cong$ a line). The two *coupled* conditions on $a+c$ and $a-c$ intersect these into a **genus-1 elliptic curve**; its rational points are the many-square magic squares.

Coordinates: scale versus shape. The two free parameters are (a, c) ; equivalently use (a, r) with

$$r = \frac{C}{A} = \frac{c}{a} \quad (\text{scale-invariant shape}), \quad a \text{ (scale)}, \quad c = ar.$$

Here r is *built from* a and c , but (a, r) is a genuine coordinate system: r is unchanged under a common rescaling $(a, c) \mapsto (\lambda a, \lambda c)$, while a carries the magnitude.

Independence of the two conditions (not of the variables).

- The *distinctness* condition (avoid the forbidden ratios) depends *only on the shape* r ; scaling a leaves it untouched.
- The *squareness* conditions depend on *both* a and r (the actual values $1 \pm a, 1 \pm ar, \dots$ must be rational squares).

Hence the two *conditions* are logically independent—choosing a permitted shape neither implies nor precludes the existence of a scale a making the cells square—so they cannot contradict each other. In particular, squaring a magnitude never forces r into the forbidden set: the record family below squares one diagonal magnitude fully at $r \notin$ forbidden set.

6 Modular sieves on the offsets

A square is 0 or 1 (mod 3), 0, 1, 4 (mod 8), etc. Writing the nine cells as $E \pm (\text{offset})$ gives necessary conditions on A, C :

$$\begin{array}{lll} \text{mod } 3 : & A \equiv C \equiv 0 & (\text{necessary for } \geq 7 \text{ squares}) \\ \text{mod } 8/16 : & A \equiv C \equiv 0 \text{ mod } 8 & \\ \text{mod } 5 \text{ (last digit)} : & A \equiv 0 \text{ or } C \equiv 0 & \\ \hline \text{mod } 240 = 16 \cdot 3 \cdot 5 : & \text{only } 1/2880 \text{ of } (A, C) \text{ survive} & \end{array}$$

Every record family passes all sieves. A pair value from $\text{squarepairs}(2E)$ is already a perfect square, hence passes every sieve automatically—so drawing free cells from square-pairs is strictly stronger than the sieve and runs in $O(\text{pairs}^2)$.

7 Eigenmatrix of $\mathbf{N}$

The offset matrix is rank 2 and generates its own powers:

$$\boxed{\mathbf{N}^2 = (a^2 - c^2)(3\mathbf{I} - \mathbf{J})}, \quad \boxed{\mathbf{N}^3 = 3(a^2 - c^2)\mathbf{N}}.$$

Writing $\kappa = a^2 - c^2 = (a + c)(a - c)$ (the product of the two diagonal magnitudes), the eigenvalues are

$$\text{spec}(\mathbf{N}) = \left\{ 0, +\sqrt{3\kappa}, -\sqrt{3\kappa} \right\}.$$

The all-ones vector $\mathbf{1}$ is the 0-eigenvector (line sums vanish). Note $\mathbf{N}^2$ has only two distinct entries—squaring collapses the shape to the single scalar κ ; the nine-distinct structure lives only in the *linear* $\mathbf{N}$.

Derivation of $\mathbf{N}^{1/2}$

Since $\mathbf{N}$ generates its own powers

$$\mathbf{N}^2 = \kappa(3\mathbf{I} - \mathbf{J}), \quad \mathbf{N}^3 = 3\kappa\mathbf{N}, \quad \mathbf{N}^4 = \mathbf{N} \cdot \mathbf{N}^3 = 3\kappa\mathbf{N}^2 = 3\kappa^2(3\mathbf{I} - \mathbf{J}), \quad \kappa = a^2 - c^2,$$

any polynomial in $\mathbf{N}$ closes on the two building blocks $\mathbf{N}$ and $(3\mathbf{I} - \mathbf{J})$. Because $\mathbf{N}$ annihilates the all-ones vector (0 eigenvalue), the square root has no constant term, so it is a *quadratic in $\mathbf{N}$* :

$$\mathbf{N}^{1/2} = q\mathbf{N} + p\mathbf{N}^2.$$

Squaring and reducing with the identities above,

$$(q\mathbf{N} + p\mathbf{N}^2)^2 = q^2\mathbf{N}^2 + 2pq\mathbf{N}^3 + p^2\mathbf{N}^4 = \underbrace{(q^2\kappa + 3p^2\kappa^2)}_{\text{coeff. of } (3\mathbf{I} - \mathbf{J})} (3\mathbf{I} - \mathbf{J}) + \underbrace{6pq\kappa}_{\text{coeff. of } \mathbf{N}} \mathbf{N}.$$

Matching this to $\mathbf{N}$ (no $(3\mathbf{I} - \mathbf{J})$ part, unit $\mathbf{N}$ part) gives the two coefficient equations

$$6pq\kappa = 1, \quad q^2\kappa + 3p^2\kappa^2 = 0 \quad \implies \quad pq = \frac{1}{6\kappa}, \quad q^2 = -3\kappa p^2.$$

Eliminating in turn,

$$\boxed{\mathbf{N}^{1/2} = q\mathbf{N} + p\mathbf{N}^2}, \quad \boxed{q^4 = \frac{-1}{12\kappa}, \quad p^4 = \frac{-1}{108\kappa^3}}, \quad \kappa = a^2 - c^2.$$

Because $p^4, q^4 < 0$, $\mathbf{N}^{1/2}$ is **complex** (forced by the eigenvalue $-\sqrt{3\kappa}$). It is *semi-magic* (rows and columns sum to 0) but not full magic (diagonals differ), and it fails to exist when $\kappa = 0$, i.e. at the forbidden diagonal ratios $r = \pm 1$ (there $\mathbf{N}$ is nilpotent, $\mathbf{N}^2 = 0$).

Limits on the coefficients p, q from the a, c limits

Everything passes through the single invariant $\kappa = a^2 - c^2 = a^2(1 - r^2)$. The shape and positivity limits of §4 pin its range:

$$0 < r < 1 \Rightarrow \kappa > 0, \quad |a| < \frac{1}{1+r} \Rightarrow \kappa < \frac{1-r}{1+r} < 1 \quad \Rightarrow \quad \boxed{0 < \kappa < 1}.$$

The supremum $\kappa \rightarrow 1$ is approached as $r \rightarrow 0$, $a \rightarrow 1$; the infimum $\kappa \rightarrow 0$ as $r \rightarrow 1$ (the degenerate diagonal). From $|p|^4 = \frac{1}{108\kappa^3}$ and $|q|^4 = \frac{1}{12\kappa}$,

$$|p| = \frac{1}{(108\kappa^3)^{1/4}}, \quad |q| = \frac{1}{(12\kappa)^{1/4}},$$

which, on $0 < \kappa < 1$, give

$$\boxed{|p| > 108^{-1/4} \approx 0.3102, \quad |q| > 12^{-1/4} \approx 0.5373} \quad (\text{both} \rightarrow \infty \text{ as } \kappa \rightarrow 0, r \rightarrow 1).$$

The two coefficients are tied by clean invariants:

$$|pq| = \frac{1}{6\kappa} > \frac{1}{6}, \quad \left| \frac{q}{p} \right| = \sqrt{3\kappa} < \sqrt{3} \quad (= |\mu|, \text{ the nonzero eigenvalue}).$$

So the square-root coefficients have *lower* bounds $(108^{-1/4}, 12^{-1/4})$ at the maximal shape ($\kappa \rightarrow 1$) and *blow up* at the degenerate diagonal ($\kappa \rightarrow 0$, $r \rightarrow \pm 1$), consistent with $\mathbf{N}^{1/2}$ ceasing to exist there. The phase of p, q is fixed by $p^4, q^4 < 0$ (fourth roots of a negative real, i.e. arguments $\frac{\pi}{4} + \frac{k\pi}{2}$).

Remark: factoring $6pq$ yields the diagonal magnitudes

Because $\kappa = (a+c)(a-c)$ and $pq = \frac{1}{6\kappa}$, the reciprocal

$$6pq = \frac{1}{\kappa} = \frac{1}{(a+c)(a-c)}$$

carries the whole diagonal structure. A single value pq fixes only the invariant κ (a hyperbola $a^2 - c^2 = \text{const}$, with the shape r still free), but a *factorization* $6pq = m \cdot n$ distributes it into the two diagonal magnitudes:

$$a+c = \frac{1}{m}, \quad a-c = \frac{1}{n} \quad \Rightarrow \quad a = \frac{m+n}{2mn}, \quad c = \frac{n-m}{2mn}, \quad r = \frac{c}{a} = \frac{n-m}{n+m}.$$

Each factor pair is a distinct rational shape. For example $pq = 35$ gives $6pq = 210 = 2 \cdot 3 \cdot 5 \cdot 7$; the pairing with the 6 is $(m, n) = (6, 35)$, whence

$$a+c = \frac{1}{6}, \quad a-c = \frac{1}{35}, \quad a = \frac{41}{420}, \quad c = \frac{29}{420}, \quad r = \frac{29}{41},$$

and the eight factor pairs of 210 give eight admissible shapes (all in $(0, 1)$, none equal to $\frac{1}{2}$). Thus the *product* pq sets the invariant, while *factoring* $6pq$ is exactly factoring $(a+c)(a-c)$ into the two diagonals.

8 Shape $\leftrightarrow$ factorization

The additive representation $a+c = \frac{1}{m}$, $a-c = \frac{1}{n}$ (with $mn = 6pq$) links the *shape* to the *factorization* through two exact, additive bridges.

The shape numerators are symmetric about n

Over the common denominator $2mn$, the offsets are

$$a = \frac{m+n}{2mn}, \quad c = \frac{n-m}{2mn},$$

so the two numerators are $n+m$ and $n-m$: symmetric about n with half-gap m ,

$$c\text{-num} = n-m, \quad a\text{-num} = n+m, \quad \text{center} = \frac{(n+m) + (n-m)}{2} = n, \quad \text{half-gap} = m.$$

The Möbius map on the shape recovers the factor pair exactly:

$$\boxed{\frac{1+r}{1-r} = \frac{n}{m}}, \quad r = \frac{c}{a} = \frac{n-m}{n+m},$$

and the reciprocal shape $r \mapsto 1/r$ flips the sign, $\frac{1+1/r}{1-1/r} = -\frac{n}{m}$. (This is the same 45° Möbius map of §5, turning axis magnitudes into diagonal ones.) In the special case $m=6$ the center is $n=pq$ itself and the half-gap is 6, e.g. $71 = 77 - 6$, $83 = 77 + 6$, giving $\frac{1+r}{1-r} = \frac{77}{6}$.

Euler: two sums of two squares factor the center

The magic-square richness lives in $2c = u^2 + v^2$; a center with two *distinct* such representations factors by Euler's method:

$$N = a^2 + b^2 = c^2 + d^2 \quad (\{a, b\} \neq \{c, d\}) \implies \gcd(ad \pm bc, N) \text{ are nontrivial factors of } N.$$

For instance $65 = 1^2 + 8^2 = 4^2 + 7^2 \Rightarrow \gcd(1 \cdot 7 \mp 8 \cdot 4, 65) = 5, 13$, and $2501 = 1^2 + 50^2 = 10^2 + 49^2 \Rightarrow 41, 61$. Hence

$$\text{many additive reps of } 2c \iff c \text{ built from primes} \equiv 1 \pmod{4} \iff \text{Euler factors } c.$$

Richness ($\text{square_pairs}(2c)$) and factorability are the same fact: this is why the record families sit at highly composite centers.

Why neither factors “for free”

Both bridges are genuine identities, yet neither is a shortcut, for the same reason:

- $\frac{1+r}{1-r} = \frac{n}{m}$ is a *bijection* shape $\leftrightarrow$ factor pair. Using it presupposes the shape $r = C/A$, which is itself built from m, n ; it returns what was put in.
- Euler's gcd is instant *given* two representations, but finding even one (searching a with $N - a^2$ square) costs $\Theta(\sqrt{N})$.

Any fixed arithmetic rule that output a nontrivial factor of N would either take a factor as (possibly hidden) input or scan $\Theta(\sqrt{N})$ candidates with a divisibility test—so no residue/floor/Möbius expression escapes the $\sqrt{N}$ cost. The additive lens *organizes* the problem correctly; it does not cheapen the factoring.

Where “peel 2, 3” legitimately leads: the $6k \pm 1$ wheel

There *is* a real, correct algorithm at the end of this line, and it is the one the $6k \pm 1$ structure points to. Consider the candidate modulus

$$M = (N \bmod 6) + 6m, \quad m = 0, 1, 2, \dots$$

For N coprime to 6 (i.e. after peeling the factors 2 and 3), $N \bmod 6 \in \{1, 5\}$, so M runs through one residue class $\{5, 11, 17, 23, \dots\}$ or $\{1, 7, 13, 19, \dots\}$ —exactly the $6k \pm 1$ numbers, the only possible prime factors of N . Testing $N \bmod M = 0$ is a divisibility check, and scanning m is **wheel** ($6k \pm 1$) **trial division**: a genuine method, $\sim 3\times$ faster than testing every integer because multiples of 2 and 3 are skipped. Two requirements make it correct, and one caveat bounds it:

- **Scan m .** A single fixed m tests one candidate; factoring needs m incremented until $M \mid N$ or $M > \sqrt{N}$.
- **Both classes.** A factor may lie in *either* $\{1, 5\} \bmod 6$. E.g. $55 = 5 \cdot 11$ has $N \equiv 1 \pmod{6}$ yet both primes are $\equiv 5$; scanning only the class $N \bmod 6$ misses them.
- **Still $\Theta(\sqrt{N})$.** The wheel removes a constant fraction of candidates but marches M up to $\sqrt{N}$; it is a constant-factor win, not an asymptotic one.

This is the honest terminus of the “peel 2, 3, then everything is $6k \pm 1$ ” structure (the same $6k \pm 1$ blocks $5 = 6 - 1$, $7 = 6 + 1$ that seed the flanking-pair arithmetic): the residue and Möbius shortcuts dissolve under a fair sample of semiprimes, but the $6k \pm 1$ wheel survives—because it *is* trial division, in its 2, 3-peeled form, and therefore respects the $\sqrt{N}$ cost rather than evading it.

Closing the loop: the $\{1, 2, 3, +, -, *\}$ canonical form

The same $6k \pm 1$ fact gives a canonical representation of *every* natural number over the alphabet $\{1, 2, 3, +, -, *\}$. Factor N with the wheel; then 2 and 3 are atoms, and every prime $p > 3$ is

$$p = 2 \cdot 3 \cdot k \pm 1, \quad k = \frac{p \mp 1}{6},$$

with k written in the same form, recursively. Thus

$$N = \prod_i p_i, \quad p_i = 2 \cdot 3 \cdot (\text{canonical } k_i) \pm 1.$$

For example

$$5 = (2 \cdot 3 \cdot 1 - 1), \quad 7 = (2 \cdot 3 \cdot 1 + 1), \quad 17 = (2 \cdot 3 \cdot 3 - 1),$$

$$425 = 5 \cdot 5 \cdot 17 = (2 \cdot 3 \cdot 1 - 1)(2 \cdot 3 \cdot 1 - 1)(2 \cdot 3 \cdot 3 - 1), \quad 180625 = 425^2 = (2 \cdot 3 \cdot 1 - 1)^4 (2 \cdot 3 \cdot 3 - 1)^2,$$

and primes recurse, e.g. $2501 = 41 \cdot 61 = (2 \cdot 3 \cdot (2 \cdot 3 \cdot 1 + 1) - 1)(2 \cdot 3 \cdot 2 \cdot (2 \cdot 3 \cdot 1 - 1) + 1)$. The wheel is the engine (it yields the primes); the $6k \pm 1$ identity is the rule that rebuilds each prime from $\{1, 2, 3, \pm 1, *\}$. This is the same single fact—*after 2 and 3, everything is $6k \pm 1$* —that underlies the flanking-pair richness ($2c = u^2 + v^2$ needs primes $\equiv 1 \pmod{4}$ but the blocks are $6k \pm 1$), the factorization wheel, and this canonical form: one structure, three faces.

9 Summary of value limits

Quantity	Condition / limit
Center E	$E = \rho^2$ (perfect square) so cell/ E is a rational square; sum = $3E$.
Degrees of freedom	2 shape (a,c) + 1 scale E ; semi-magic has 4.
Shape ratio $r = C/A$	fundamental domain $0 < r < 1$; symmetry $r \sim \{-r, 1/r, -1/r\}$.
Forbidden r (no 9 distinct)	$r \in \{0, \pm\frac{1}{2}, \pm 1, \pm 2, \infty\}$ (only $\frac{1}{2}$ inside $(0, 1)$).
Offset scale a (positivity)	$ a < \frac{1}{1+r}$ for $0 < r < 1$.
Offsets A, C (mod sieve, ≥ 7 sq.)	$A \equiv C \equiv 0 \pmod{3}$; combined mod 240 keeps 1/2880.
Eigenvalues of $\mathbf{N}$	$0, \pm\sqrt{3(a^2 - c^2)}$; real iff $ r < 1$.
Invariant κ	$\kappa = a^2 - c^2 = (a+c)(a-c)$; $\kappa > 0$ on $0 < r < 1$; $\kappa = 0$ at $r = \pm 1$.
Cells vs. sum	every cell $< 2E = \frac{2}{3}S$ (opposite > 0).

10 Worked examples: the 425 and 195364 squares

In each grid the nine cells are shown with perfect squares marked * (root given in the caption). The right-hand column lists the three **row sums**; the bottom row lists the three **column sums**; and the bottom-right corner gives the two **diagonal sums** ($\backslash$ = main, $/$ = anti).

139129*	83521*	319225*	541875
360721	180625*	529*	541875
42025*	277729*	222121	541875
541875	541875	541875	$\backslash$ 541875 $/$ 541875

Table 1: 425 **family** — root $\rho = 425$, center $E = 425^2 = 180625$, magic sum $3E = 541875$. All rows, columns *and* both diagonals equal 541875. **Magic**, 7 squares: $373^2, 289^2, 565^2, 425^2, 23^2, 205^2, 527^2$; the two non-squares are 360721 and 222121.

40804*	386884*	158404*	586092
312964	195364*	77764	586092
232324*	3844*	349924	586092
586092	586092	586092	$\backslash$ 586092 $/$ 586092

Table 2: 195364 **magic square** — center $E = 442^2 = 195364$, magic sum $3E = 586092$. All rows, columns and both diagonals equal 586092. **Magic**, 6 squares: $202^2, 622^2, 398^2, 442^2, 482^2, 62^2$; non-squares 312964, 77764, 349924. This is the best *magic* square-count reachable at this center.

3844*	33124*	56644*	93612
158404*	195364*	232324*	586092
334084*	357604*	386884*	1078572
			\\ 586092
496332	586092	675852	/ 586092

Table 3: 195364 **nine-square** (point-symmetric) — all 9 cells are squares ($62^2, 182^2, 238^2, 398^2, 442^2, 482^2, 578^2, 598^2, 622^2$), but **NOT magic**. Only the center cross reaches 586092: middle row, middle column, and *both diagonals* ($\backslash = / = 586092$, shown in the corner). The outer rows (93612, 1078572) and outer columns (496332, 675852) are off by $\pm X$ ($X = 492480$ for rows) — the signature of a point-symmetric grid that is not magic.

349924	33124*	386884*	769932
232324*	195364*	158404*	586092
3844*	357604*	40804*	402252
			\\ 586092
586092	586092	586092	/ 586092

Table 4: 195364 **eight-square** (point-symmetric) — 8 of the 9 cells are squares ($182^2, 622^2, 482^2, 442^2, 398^2, 62^2, 598^2, 202^2$); only the top-left 349924 is not ($\sqrt{349924} \approx 591.54$). Here *all three columns* and *both diagonals and* the middle row equal 586092 — six of the eight lines are magic — but the top and bottom rows are off by ± 183840 (769932 and 402252). It is the closest approach to a magic 8-square: to make it magic the top-left cell is forced to $586092 - 33124 - 386884 = 166084 = 407.5^2 \dots$, which is *not* a perfect square. That single stubborn cell is the $7 \rightarrow 8$ wall in one picture.

11 Records and open problem

- **9 distinct squares, non-magic:** easy—any center with ≥ 4 flanking square-pairs (point-symmetric grid).
- **Magic, 7 squares:** the “425 family” at root $\rho = 425$, center $E = 180625$, shape $r = -247/825$ (in $(0, 1)$ up to symmetry, avoids $\frac{1}{2}$). Two of the four magnitudes are fully squared.
- **Magic, 8 or 9 squares:** *open*. Requires all four magnitudes $1 \pm \{a, c, a + c, a - c\}$ simultaneously square—a rational point on the elliptic curve. No local/modular obstruction rules out any admissible r ; the obstruction, if any, is global.

Acknowledgments

This note grew out of an extended exploration by the author, reducing the 3×3 magic square of squares to its two-parameter shape (a, c) , the single scale-invariant ratio $r = C/A$, and the offset matrix $\mathbf{N} = a\mathbf{P} + c\mathbf{Q}$. The symbolic identities ($\mathbf{N}^2 = (a^2 - c^2)(3\mathbf{I} - \mathbf{J})$, $\mathbf{N}^3 = 3(a^2 - c^2)\mathbf{N}$, the closed form of $\mathbf{N}^{1/2}$), the modular sieves, and the numerical checks throughout were verified computationally with SymPy and NumPy. The author thanks these open-source tools, whose symbolic and numerical support made the verification of every claim in this note routine.